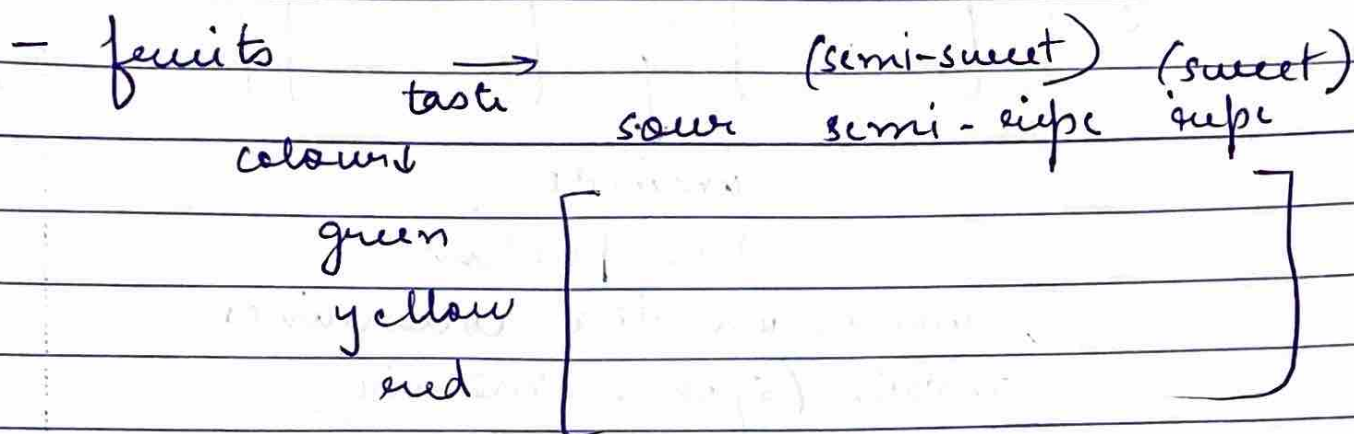


5<sup>th</sup> March 2026 / CS6370

## ⇒ LSA (Latent Semantic Analysis)



- ripeness actually decides the colour of the fruit. As we go towards the root cause, the problem of circularity tends to disappear.

- latent variables → if we can represent both the features to the latent variables (underlying concepts) then this circularity problem has been resolved.

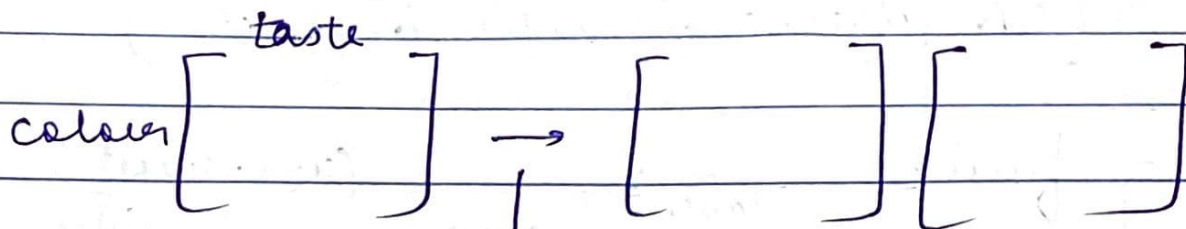
- They have an explanatory value, which sth is happening.

$$\text{ripe} = \{0.1 \cdot \text{green} + 0.3 \cdot \text{yellow} + 0.6 \cdot \text{red}\}$$

$$\text{ripe} = \{\text{linear combination of taste}\}$$

express  $c_1, c_2, \& c_3$  in terms of colours  
latent variables

- matrix decomposition

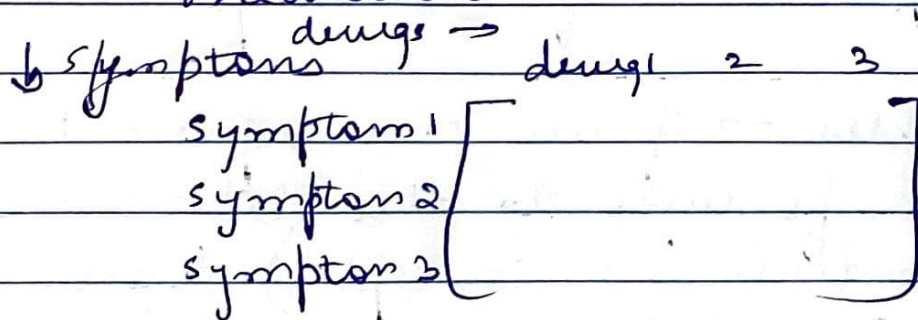


unimodal  
decomposition

works on the covariance  
matrix (square matrix)

we are moving from taste space to  
colour space.

- medicines



- latent factors is a conceptual reason.

- no. of latent variables must be much  
lesser than actual observations.

m rows

n columns

k latent var.

$$k(m+n) \ll mn$$